

Name: AMRITANSU SINGH AMRITANSU SINGH | Reg. No: RA2511003011028

Unit 5 : Session 6 : SLO 1 | Symbolic Calculus Differentiation and Integration

I. Questions & Answers

1. Differentiate x^3 $\frac{d}{dx}(x^3) = 3x^2$. In SymPy: `diff(x**3, x)` $\rightarrow 3x^2$.
2. Integrate $2x$ $\int 2x dx = x^2 + C$. In SymPy: `integrate(2*x, x)` $\rightarrow x^2$ (constant omitted).
3. Differentiate $\sin(x)$ $\frac{d}{dx} \sin(x) = \cos(x)$. In SymPy: `diff(sin(x), x)` $\rightarrow \cos(x)$.

II. Code

- `from sympy import symbols, diff, integrate, sin`
- `x = symbols('x')`
- `print(diff(x**3, x))` $\rightarrow 3x^2$
- `print(integrate(2*x, x))` $\rightarrow x^2$
- `print(diff(sin(x), x))` $\rightarrow \cos(x)$